INCH-POUND

MIL-V-80240C
15 June 1990
SUPERSEDING
MIL-V-80240B
6 July 1977

MILITARY SPECIFICATION

VIEWERS, MICROFICHE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for militarized, dual-lens, microfiche viewers capable of enlarging 24:1 and 48:1 microimages recorded on a standard 4.13 by 5.83 inches (in.) (105 by 148 millimeter (mm)) microfiche and projecting the images onto self-contained screens and capable of accommodating, enlarging and displaying segments of microimages recorded on 3.26 by 7.38 in. (82.73 by 187.45 mm) aperture cards (see 6.5.9, 6.5.11, and 6.5.19).

1.2 Classification. Unless otherwise specified (see 6.2), microfiche viewers covered by this specification are classified by size, by direction of image projection on viewing screen, and by lens magnification (see 6.5.6, 6.5.7, and 6.5.18).

1.2.1 Size. Unless otherwise specified, viewers are classified in the following sizes:

SIZE 1 - Size 1 (full size) viewers are capable of enlarging 24:1 and 48:1 microimages and displaying them at essentially 100 percent of the original document size. These viewers have a usable screen size that is not less than 11.0 in. high and 14.0 in. wide. Size 1 viewers will pass through a 25 in. diameter round hatch.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense General Supply Center, DGSC-SSC, Richmond, VA 23297-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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the defects in table I. The inspection level shall be [illegible] II with [illegible]

SIZE 2 - Size 2 (three-quarter size) viewers are capable of enlarging 24:1 and 48:1 microimages and displaying them at approximately 75 percent of the original document size. Size 2 viewers have a usable screen size that is not less than 8.25 in. high and 10.5 in. wide. A size 2 viewer can be installed and will properly operate in an open-front, rectangular cavity having dimensions of 19.0 in. high, 14.0 in. wide, and 15.25 in. deep.

1.2.2 Projection. Viewers in which the enlarged image is projected on the front of the screen, as seen from the operator's position, are classified as front projection viewers. Viewers in which the enlarged image is projected on the rear of the screen, as seen from the operator's position, are classified as rear projection viewers. The following projection codes designate the direction of a viewer's image projection.

F - Front projection - image projected on front of viewing screen.
R - Rear projection - image projected on rear of viewing screen.

1.2.3 Lenses. Viewers are procured with a set of two lenses. The following lens codes indicate the magnifications of the lenses that are included with the identified item (see 3.3.11). The normal usage of these lens sets is indicated in Table I.

18X36X - consists of one 18X lens and one 36X lens.
24X48X - consists of one 24X lens and one 48X lens.

1.3 Part or identifying number (PIN). The microfiche viewers covered by this specification are identified by the specification-based part or identifying number system described in 6.7. The "standard" viewers available under this specification are listed in Table I.

Table I. Standard viewers.

Part number	Description
M80240-1F24X48X	Full-size, front projection viewer with 24X and 48X lenses.
M80240-1R24X48X	Full-size, rear projection viewer with 24X and 48X lenses.
M80240-2R18X36X	Three-quarter-size, rear projection, viewer with 18X and 36X lenses.

2. APPLICABLE DOCUMENTS

2.1 Government specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to

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the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-C-596/13	-	Connector, Plug, Electrical, General Purpose, Hospital Grade, Grounding, 2 Pole 3 Wire, 15 Amperes, 125 volts, 50/60 Hertz
PPP-B-636	-	Box, Shipping, Fiberboard
PPP-T-60	-	Tape, Packaging, Waterproof
PPP-T-76	-	Tape, Pressure-Sensitive, Adhesive, Paper, (for Carton Sealing)

MILITARY

MIL-P-116	-	Preservation-Packaging, Method of
MIL-C-675	-	Coating of Glass Optical Elements (anti-reflection)
MIL-M-7298	-	Manual, Technical, Commercial Equipment
MIL-C-28777	-	Cable Assembly, Electronic Test Equipment, (3 wires, 125 and 250 volts AC and 28 volts DC), Grounding Plug Connector, General Specification for
MIL-C-28777/2	-	Cable Assembly, (Type I), Electronic Test Equipment, (2 poles, 3 wires, 125 volts, 13 amperes, 50 or 60 Hertz), Grounding Plug Connector and Terminal Lugs
MIL-C-28777/5	-	Cable Assembly, (Type I), Electronic Test Equipment, (2 poles, 3 wires, 125 volts, 13 amperes, 50 or 60 Hertz), Grounding Plug Connector

STANDARDS

FEDERAL

FED-STD-376	-	Preferred Metric Units for General Use by the Federal Government

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-147	-	Palletized and Containerized Unit Loads for 40" x 48" Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-167-1(Ships)	-	Mechanical Vibrations of Shipboard Equipment
MIL-STD-399	-	Microform Formats

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MIL-STD-454 - Standard General Requirements for Electrical Equipment
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810 - Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications and standards required are available from the DoD Single Stock Point at the following address: Standardization Documents Order Desk, Building 4D, 700 Robins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues of documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. The issues of documents not listed in the DODISS are the issues cited in the solicitation (see 6.2).

INDUSTRY STANDARDS

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI PH3.46-1971 - Projected-media Temperature Test for Photographic Still-Projection Equipment
ANSI/NMA MS-12-1977 - Method for Measuring the Screen Luminance of Microfilm Readers with Translucent Screens
ANSI S1.4-1971 - General Purpose Sound Level Meters, Specification for
ANSI X4.12-1970 - Office Appliances and Business Equipment, Safety Standard for (UL 114 October 1970)

(ANSI Standards may be obtained from the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries and other informational sources.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exception has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

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3.2 Material (see 4.9.1). Materials used in the fabrication of parts for viewers shall be as specified herein. However, when definite materials or material requirements are not specified, materials shall be used which will enable viewers to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.

3.2.1 Breakage resistant. The screen, shell, and other exterior parts and the mechanical and structural parts of viewers shall be made of materials that are shatter resistant and are not easily broken or bent.

3.2.2 Flammability. Viewers shall be constructed of materials that are not flammable, will not support combustion, and will, to the maximum extent possible, produce a minimum of toxic and noxious gases when burned or thermally decomposed. Viewer shall conform to the requirements of MIL-STD-454, Requirement 3.

3.2.3 Dissimilar metals. Direct contact of dissimilar metals shall be avoided wherever practical. Where contact is necessary, adequate protection against electrolytic corrosive action shall be provided. Dissimilar metal combinations shall conform to the requirements of MIL-STD-454, Requirement 16.

3.2.4 Fungus-inert materials. Viewers shall conform to the requirements of MIL-STD-454, Requirement 4, including paragraph 3.3, Carcinogens. For purposes of MIL-STD-454, Requirement 4, viewers shall be considered as new design.

3.2.5 Corrosion-resistant ferrous alloys. The finishes of corrosion-resistant ferrous alloys shall conform to the requirement of MIL-STD-454, Requirement 15.

3.3 Design and construction (see 4.9.2). Viewers shall be designed and constructed to illuminate, enlarge, and project 24:1 and 48:1 microimages on self-contained viewing screens. Viewers shall be designed and constructed to withstand normal usage, including usage by an inexperienced user, transport, routine handling, and the environmental conditions that exist wherever they may be used by the military establishment worldwide. Viewers shall be designed for desk-top usage. They shall include base pads or mountings which will prevent damage to furniture or other surfaces on which the viewer may be placed and which will provide non-slip capability necessary to prevent undesirable movement of an unsecured viewer as a result of normal usage.

3.3.1 General. Viewers shall be sufficiently rigid to prevent the misalignment of optical components or damage to viewer by normal handling and/or usage. The shells of viewers may be used as a support member for internal mechanisms. The power switch and all other operator controls shall be located on the front side of viewers (the side facing the operator) below

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the operator's line of vision to the screen and shall conform to the provisions of MIL-STD-454, Requirements 28 and 48. Appropriate warning signs shall be installed at strategic points on viewers indicating the presence of hazardous conditions such as hot spots, voltages in excess of 30 volts (v) RMS (42.4 v peak), etc. Requirement 1 of MIL-STD-454 applies.

3.3.2 Securing device. A means of securing viewers to horizontal metal and wooden surfaces shall be provided. All accessories required for securing a viewer shall be included with the viewer. For type 2 viewers, controls for the attachment mechanism shall be at the front of the viewers and the use of the attachment mechanism shall not require space larger than the cavity defined in 3.3.5.2. The viewers shall be fully operable and meet all other requirements whether or not the securing device is used.

3.3.3 Primers/enamels. All metallic surfaces which are not corrosion resistant shall be painted. When painted, metallic surfaces shall be prepared for enamel by being covered with one or more coats of metallic base primer or by phosphatizing the metal in lieu of primer, and with one or more coats of top quality synthetic enamel. Primer and finish coats shall be applied in accordance with normal industrial painting practice. The finish coats when dry shall present a smooth, even surface free from runs, sags, peels, chips, blisters, and scratches and be free of thin-coated or non-coated areas. The colors of the enamel shall be sufficiently different from color of the primer or phosphatized layer that thin or non-coated areas in the enamel coat will be easily discerned.

3.3.4 Finished condition. All exterior surfaces of viewers shall be dry and free from oil, grease, dirt, burrs, sharp edges, and rust. Viewer shall be clean, well made, and free from defects which affect their serviceability and/or detract from their appearance.

3.3.5 Viewer size.

3.3.5.1 Size 1. The overall dimensions of the size 1 viewer shall be such as to permit passage of a viewer through a circular hatch 25.0 in. in diameter. The size 1 viewer may exceed these dimensions if it is capable of being readily disassembled by nontechnical personnel into a maximum of four parts (excluding hoods and any hardware used in holding the viewer parts together, such as nuts, bolts, washers, etc.), each of which will pass through the hatch, and be readily reassembled by nontechnical personnel, in which case appropriate instructions for such disassembly and reassembly shall be provided with each viewer in the technical manuals. The total viewer weight shall not exceed 50 pounds.

3.3.5.2 Size 2. The overall dimensions of the size 2 viewer, excluding hoods, shall not exceed 18.5 in. in height, 13.5 in. in width, and 15.0 in. in depth with the carrier in the stowed position. When placed in an open-front rectangular cavity not exceeding 19.0 in. high, 14.0 in. wide, and 15.25 in.

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deep, the size 2 viewer shall be fully operational and usable. The carrier may extend a not more than 6 in. in front of the cavity for inserting, viewing, and removing microfiche. Hoods, when used, may extend not more than 6 in. in front of the cavity. The total viewer weight shall not exceed 35 pounds.

3.3.6 Viewing screen. Each viewer shall be equipped with a non-glare viewing screen capable of producing an image with good contrast and legibility. The top of the apparent screen (see 6.5.2) shall not be tilted from vertical by more than 5 degrees toward the operator or 20 degrees away from the operator. The usable viewing area of the apparent screen of a size 1 viewers shall be not less than 11.0 in. high and 14.0 in. wide. The usable viewing area of the apparent screen of a size 2 viewers shall be not less than 8.25 in. high and 10.5 in. wide. All non-optical surfaces between the front of front projection viewers and the viewing screens shall have a dark, non-reflective finish.

3.3.6.1 Screen design. The viewing surface and any transparent shield shall be free of scintillation and be such that the reflection of the ambient lighting (see 3.4.1 and 6.5.1) shall not interfere with the legibility of the image. The viewed image shall be clear and legible to an operator whose eyes are within the viewing zone (see 6.5.20).

3.3.6.2 Impact and scratch resistance. The viewing screen and any transparent shield shall be shatter-resistant and their surfaces shall be sufficiently scratch resistant to withstand, without damage, normal cleaning in accordance with procedures specified by the manufacturer. The manufacturer's cleaning procedure for the screen and a suitable cleaning procedure for any transparent shield shall be included as part of the operation instructions in the technical manuals.

3.3.6.3 Ease of removal. The viewing screen on rear projection viewers and any transparent shield on front or rear projection viewers shall be designed to be removed and replaced in less than 5 minutes.

3.3.6.4 Marker line. To facilitate retrieval of data from a single line, viewers shall have a transparent 0.118 + 0.039 in. horizontal marker line on the viewing screen at its approximate center and extending across the entire viewing area. The marker line shall be prominent and easily distinguished from projected images which have been enlarged from positive and negative silver, vesicular, and blue, blue-black, and neutral black diazo films. The marker line shall not obscure any part of an image on the viewing screen.

3.3.7 Hood. All rear projection viewers shall be equipped with a fixed or detachable hood, consisting of a top and two sides, as an ambient light shield. The top of the hood for size 1 viewers shall extend toward the operator not less than 6.0 in. in front of the front-most point of the screen.

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The top of the hood for size 2 viewers shall extend toward the operator not less than 4.5 in. in front of the front-most point on the screen. The sides of the hoods shall be vertical and tapered from the outer edge of the hood top to points approximately level with and 1 in. in front of the lower edge of the screen. The hood shall not block the line of sight perpendicular to any point on the screen. The interior of hoods shall have a dark, non-reflective finish.

3.3.8 Lamp. In order to prolong lumen maintenance, projection lamps in viewers shall be low voltage regenerative-cycle types, such as tungsten-halogen. Lamps shall have a rated life of not less than 500 hours when operated in the viewers. The lamp voltage shall not exceed 35.36 v RMS (50 v peak). The lamp shall be replaceable within 5 minutes exclusive of cooling time. Any special handling requirements for lamps shall be clearly identified on the viewers and in the technical manuals.

3.3.9 Thermal design. Viewers shall comply with the requirements of MIL-STD-454, Requirement 52. If forced air cooling blowers are used, the blowers shall have sealed, lifetime-lubricated bearings and be essentially maintenance free. The discharge vent must direct the exhaust air away from the normal position of an operator's face. For this purpose, the normal position of an operator's face shall be considered to be an 90 degree wide conic zone that is in a similar location and orientation as the viewing zone (At 26 in. away from the apparent screen, this conic zone will describe a 52 in. diameter circular disk that is parallel to and concentric with the apparent screen.).

3.3.10 Optical system (see 6.5.12). The image on the viewing surface of viewers shall be right reading and free of vibration, distortion, spherical aberration and chromatic aberration to the extent required to meet the requirements specified herein. No flicker shall be observed during operation of the viewer. Misalignment of the viewer optical path shall not occur as a result of normal usage, moving, transport, or handling of the viewer within the service conditions specified herein.

3.3.10.1 Optical system accessibility and replacement. Lenses, mirrors, and other optical components shall be easily accessible so cleaning of exposed surfaces can be accomplished within ten minutes and replacement of optical system components can be accomplished within thirty minutes (exclusive of cooling time). Optical system components shall be mounted such that realignment is not required after cleaning or replacement when reasonable care is taken. The design of size 2 viewers shall permit cleaning and replacement of optical system components while the viewers are secured within the cavity identified in 3.3.5.2. A procedure for cleaning optical system components shall be included as part of the operation instructions in the technical manuals.

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3.3.10.2 Optical system schematic. A schematic diagram of the optical system shall be affixed to the interior of the viewer in a location that is accessible during maintenance and shall be included in the technical manuals.

3.3.10.3 Mirrors. The mirrors used in the optical path shall be high quality front surface mirrors that will not introduce any distortion into the projected image.

3.3.11 Lenses. The lenses used in viewers shall be corrected for spherical aberration, coma, astigmatism, and distortion to the extent necessary to meet the requirements specified herein. The surfaces of the lens that are in contact with air shall be coated with antireflective material in accordance with MIL-C-675. Unless otherwise specified (see 1.2 and 6.2), two lenses, as identified below, shall be provided with each viewer.

3.3.11.1 Size 1 viewer lenses. The size 1 viewer shall be provide with one 24X and one 48X lens which will provide linear magnification of 22X to 24X and 46X to 48X respectively. When using the 24X lens, size 1 viewers shall display on the viewing screen not less than one complete frame of a 98-frame, MIL-STD-399 microfiche (see 6.5.4). When using the 48X lens, size 1 viewers shall display on the viewing screen not less than one complete frame of a 270-frame, MIL-STD-399 microfiche.

3.3.11.2 Size 2 viewer lenses. Size 2 viewers shall be provide with one 18X and one 36X lens which will provide linear magnification of 16.5X to 18X and 34.5X to 36X respectively. When using the 18X lens, size 2 viewers shall display on the viewing screen not less than one complete frame of a 98-frame, MIL-STD-399 microfiche. When using the 36X lens, size 2 viewers shall display on the viewing screen not less than one complete frame of a 270-frame, MIL-STD-399 microfiche.

3.3.11.3 Lens marking. As a minimum, the magnification of each lens shall be marked on its barrel in a permanent and highly legible manner. Lens focal length, lens identification data, and other appropriate information may also be marked on the lens barrel.

3.3.11.4 Lens change mechanism. Viewers shall have manually-operated means for quickly shifting between two installed lenses. Prominent markings for the shifting control shall show which lens magnification is in position for use. The movement of the microform carrier (see 3.3.16) and the platens (when used) shall not cause movement of the lens that is positioned in the optical path for use. The lens that is not positioned for use shall not interfere with the movement of the microform carrier or the platens (when used). The lenses shall be affixed in the viewers to lessen the possibility of loss and to prevent their dropping out should the viewer be inverted. Unless otherwise specified (see 6.2), partial disassembly of viewers for lens removal is allowed.

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3.3.12 Focus. The range of focus on viewers shall be sufficient to provide a sharp legible image from either side of a 0.002 to 0.009 in. thick 24:1 or 48:1 microfiche. The focus control shall permit defocusing on either side of the sharpest focus. When the viewer image is in focus at screen center, the four corners and midpoints of the four edges of the image also shall be in focus. The image shall remain in focus when microfiche is moved from frame to frame including from corner to corner. The projected image shall be of flat field sufficient to meet the required resolution, without refocusing. The projected image shall not have a keystone, trapezoid, curved, bowed, or other distorted appearance.

3.3.12.1 Focus mechanism. Focus mechanism on viewers shall be designed to permit convenient operator focusing, to be essentially free of backlash, and to remain in a set location until changed by the operator.

3.3.13 Resolution of fine detail (see 6.5.16). Viewers shall be capable of resolving the required pattern when focused on test microimages on appropriate resolution test microfiche (see 6.5.17 and 4.9.3). The resolution at any point on the viewer screen is to be read as the smallest resolution test pattern in which an observer, close to the screen, can resolve (distinguish both radial and tangential lines in the projected image). When a 18X to 24X lens is used with a 24:1 resolution test microfiche, the 3.6 chart shall resolve. When a 36X to 48X lens is used with a 48:1 resolution test microfiche, the 3.2 chart shall resolve.

3.3.14 Viewing screen brightness. At full brightness, the luminance of the center of the viewing screen of viewers shall be not less than 50 footlamberts (171.31 candela/square meter). The luminance at each of the four corners A, B, C, and D, as defined in ANSI/NMA MS12-1977, shall not be less than 20 percent of the luminance at the center of the screen (see 4.9.4).

3.3.14.1 Viewer brightness control. A control shall be provided which permits the operator to change the brightness of the display between full brightness and a 20 to 35 percent decrease in brightness, to compensate for different film optical densities, ambient light conditions, and the operator's preference. No flicker shall be observed during operation of viewers in either brightness level. The brightness control may be combined with the power switch.

3.3.15 Viewing screen image contrast. The viewing surface shall have an image contrast ratio not less than 10:1, as defined by the test procedures of ANSI/NMA MS12-1977 (see 4.9.4).

3.3.16 Microform carrier. Each viewer shall be equipped with a vibration-free carrier for retaining 0.002 to 0.009 in. thick film precisely in an optical plane (see 6.5.10).

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3.3.16.1 Carrier design. Carriers shall accommodate 4.13 by 5.83 in. microfiche and 3.26 by 7.38 in. aperture cards, as specified in MIL-STD-399. Carrier mechanisms that require opening and closing for insertion or removal of the microforms shall be self-opening and remain open when pulled toward the operator at any left-to-right carrier position and self-closing when pushed in away from the operator. These operations shall open the front of the carrier mechanisms a minimum of 1.5 in. with a pull not exceeding 2 pounds and close the mechanisms with a push not exceeding 2 pounds. The carrier shall provide means for positive alignment of microfiche and aperture cards to minimize canted images on the viewing surface and to align the microfiche with the carrier indexing system. Carrier mechanisms shall be designed for convenient and easy operation and, once positioned, to remain in place until moved by the operator. Viewers shall provide a means to prevent accidental removal of carriers and platens (when used).

3.3.16.1.1 Platens. When platens are used, a means shall be provided for loading and unloading microforms without scratching or otherwise damaging the microforms. The ends of an aperture card may protrude outside platens or other carrier mechanisms but the card shall not be bent when inserted in the carrier or between the platens or when a carrier containing an aperture card is moved to the extreme limits of its travel. All surfaces of any platens shall be easily accessible for cleaning and shall be scratch resistant. Platens shall be easily replaceable by one person.

3.3.16.3 Carrier latch. A latch or other retaining method shall be provided to immobilize the carrier in a closed position when not in use to prevent movement and potential damage when the viewer is moved or transported or subjected to conditions such as severe roll and pitch, vibration, or shock.

3.3.16.4 Carrier indexing system. A carrier indexing system shall be provided to assist operators in positioning specific frames of both 98-frame and 270-frame microfiche in the optical path of the viewer. A pointer on the front of the carrier may be used with a fixed grid (see 6.5.5) index, indicating microimage locations by row and column, to position a selected frame of the microfiche in the optical path of the viewer. When used, index plates shall be durable and scratch resistant and, if the index plates for the 98-frame and the 270-frame microfiche are separate parts, they shall be exchangeable within 30 seconds and a means of storage, accessible from the front of the viewer, shall be provided for the index that is not mounted for use.

3.3.16.5 Carrier durability. The viewer shall remain operational and shall meet the requirements of 3.3.16 through 3.3.16.4 and 3.6 after the carrier has been cycled from left to right to left, from front to back to front, and from load to closed to load positions for 8000 cycles each (see 4.9.5).

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3.3.17 Electrical. Viewers shall operate satisfactorily when powered from each of the following single phase power sources: 115 + 11.5 v at 60 + 6 hertz (Hz), 115 + 11.5 v at 50 + 5 Hz, 230 + 23 v at 60 + 6 Hz, and 230 + 23 v at 50 + 5 Hz. No soldering or desoldering shall be required to change from one voltage or frequency to the other. Unless otherwise specified (see 6.2), all viewers shall be delivered ready for operation on 115 v and 60 Hz and shall be marked in a prominent location to indicate the voltage and frequency for which they are wired. All electrical mechanisms, wiring, and connections shall be enclosed, but shall be readily accessible for adjustments and maintenance. All external viewer surfaces shall be free of electrical shock hazards. Viewer shall conform to the applicable requirements of MIL-STD-454, Requirements 1 and 20.

3.3.17.1 Power switch. A power switch shall control the electrical power to the viewer lamp and shall be prominently marked. In the OFF position, the switch shall interrupt both electrical power leads (hot and neutral). Cooling blowers may be connected so that they will continue to run after the lamp is turned off and automatically stop running after the viewer has cooled. Switches shall conform to the requirements of MIL-STD-454, Requirement 58.

3.3.17.2 Electrical overload protection. Electrical circuits in each viewer shall be suitably protected by multi-pole circuit breakers or fuses to prevent damage due to electrical power surges or overloads. A fuse or one pole of a circuit breaker shall be installed in each of the incoming power leads (hot and neutral). Circuit breakers, when used, must interrupt both incoming electrical power leads when a trip condition occurs on either lead. The current rating of circuit breakers and fuses shall be properly sized for the current required by the viewer and shall not exceed 13 amperes. The current rating for fuses shall be marked adjacent to the fuse holders. The fuses shall be easily accessible and replaceable within five minutes and, for the size 2 viewer, while it remains securely mounted within the cavity described in 3.3.5.2. Circuit breakers must be manually resettable and be easily accessible to the operator when the viewer is sitting in its normal operating position. Viewers shall conform to the applicable requirements of MIL-STD-454, Requirements 8, 37, and 38.

3.3.17.3 Interlocks. Interlocks shall be incorporated to eliminate, to the greatest extent possible, the possibility of shock to operating and maintenance personnel. Interlock(s) shall disconnect all power to the viewer when the shell is opened for servicing or for changing lamps and fuses. Interlocks are not required when the electrical circuitry that is exposed does not exceed 30.0 v RMS (42.4 v peak) (see ANSI X4.12-1970). Viewers shall conform to the applicable requirements of MIL-STD-454, Requirement 1.

3.3.17.4 Power cord. A M28777/2-1 or a M28777/5-1 power cord, conforming to MIL-C-28777 and MIL-C-28777/2 or MIL-C-28777/5, shall be permanently attached to the viewer with a suitable strain relief. The plug

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molding material shall be the same as the cable jacket material. In lieu of a MIL-C-28777 power cord, the power cord may be a 96 + 3 inch long type SJ cord, having three AWG16 conductors color coded black (hot), white (neutral), and green or green with yellow stripe (safety ground), having an integrally molded plug conforming to MIL-C-28777, figure 1, or a separately attached plug conforming to W-C-596/13, and having Underwriters Laboratory approval. The outside diameter of the finished power cord shall be between 0.325 and 0.575 inches in order that plugs conforming to W-C-596/13 (115 volts) or W-C-596/17 (230 volts) may be properly attached when needed. When specified (see 6.2), the power cord shall conform to the above requirements in all respects except the plug configuration shall conform to MIL-C-28777, figure 2.

3.3.17.5 Electrical schematic. A schematic diagram of the electrical circuitry shall be affixed to the interior of the viewer in a location that is accessible during maintenance and shall be included in the technical manuals.

3.3.18 Acoustic noise. Acoustic noise generated by viewers shall not exceed 60 dB above the threshold of hearing (0 dB) on the A scale of a sound level meter conforming to ANSI S1.4-1971 (see 4.9.6).

3.3.19 Electromagnetic interference (EMI). Viewers shall conform to the EMI requirements for Visual Displays installed in Surface Ships (Class A4) as described in MIL-STD-461. When viewers are tested as specified in 4.9.7, their susceptibility to electromagnetic energy and the electromagnetic energy leaving viewers by radiation or conduction shall not exceed the specified levels. The EMI requirements shall not apply to the manual operation of the power switch or the brightness control.

3.3.20 Shock. Viewers shall be designed and constructed to withstand shocks expected to be encountered in normal military usage and transport. After being tested as specified in 4.9.8, viewers shall meet the requirements of 3.6.

3.3.21 Vibration. Viewers shall designed and constructed to withstand mechanical vibration of shipboard installation. During and after testing as specified in 4.9.9, viewers shall meet the requirements of 3.6.

3.3.22 Collapsibility. Viewers designed to be collapsible shall remain mechanically operative and shall meet the requirements of 3.3.16 through 3.3.16.4 and 3.6 after they have been erected, collapsed, and erected 4000 times (see 4.9.10).

3.4 Operating environment. Viewers shall meet the requirements of this specification when operating in the following environmental conditions.

3.4.1 Ambient lighting. Viewers shall be satisfactorily usable under artificial white light when the illumination level is not less than 90 footcandles (968.4 lumen/meter) at desktop level.

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3.4.2 Temperature. Viewers shall operate satisfactorily in temperatures from 35 to 105 degrees Fahrenheit (F) (see 4.9.11).

3.4.3 Humidity. Viewers shall operate satisfactorily in ambient relative humidity up to 95 percent and shall meet test requirements specified in 4.9.12. During and after the tests, viewers shall meet the requirements of 3.6.

3.5 Film plane temperature. When viewers are tested as specified in 4.9.13, there shall be no microform damage or distortion.

3.6 Viewer operating test. When viewers are tested as specified in 4.9.14, each control shall function properly and the microfiche test image projected on the viewing screen shall be easily read.

3.7 Marking. Viewers shall be identified in accordance with the MIL-STD-130. The identification plate shall be located on the viewer shell and shall contain the following information:

(a) Item name
(b) Serial number
(c) Procurement instrument identification number (Contract number, purchase order number, etc)
(d) Operating voltages and frequencies
(e) Government Part or Identification Number (PIN) (see 1.3)
(f) National stock number
(g) Manufacturer's Commercial and Government Entity (CAGE) Code (formerly FSCM), name or trademark
(h) Government ownership designation symbol [US]

3.8 Technical manuals. When technical manuals are specified (see 6.2 and 6.3), two copies of each manual shall be packaged with each unit.

3.9 Part availability and identification. Lamps and fuses used in viewers shall be readily available from commercial sources. Parts shall be identified by the actual manufacturer's name and part number.

3.10 Spare parts. One spare lamp and one complete set of spare fuses (when used) shall be furnished with each viewer. A storage mount for one spare lamp shall be provided within each viewer in a location that is accessible when the viewers are opened for lamp replacement. One spare fuse for each operating fuse shall be stored within the viewer in holders in the vicinity of the respective operating fuse holders. Spare fuses shall be identified as such. When stored in viewers in storage locations provided, spare lamps and spare fuses shall be protected from deterioration and accidental damage and shall not interfere with the normal operation of the viewer.

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3.11 Special tools. No special tools shall be required for the operation, installation, servicing, or repair of viewers including the securing devices. Special tools include any tools, jigs, fixtures, stands, and templates that are not listed in the Federal Supply Catalog.

3.12 Workmanship. Viewers shall conform to the requirements of MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified (see 6.2), the supplier is responsible for the performance of all inspection requirements as specified herein and, unless otherwise specified (see 6.2) or disapproved by the Government, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in this specification.

4.1.1 Responsibility for conformance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) First article inspection (see 4.5)
(b) Quality conformance inspection (see 4.6)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed at room ambient temperature and humidity and with an electrical input power of 115 + 10 v and 60 + 6 Hz.

4.4 First article inspection (see 3.1 and 6.4). First article inspection shall consist of the following as identified in Table II:

(a) Material inspections.
(b) Visual and mechanical examinations.
(c) Operating environment tests.
(d) Electrical and mechanical tests.

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Table II. First article and quality conformance inspections. (1)

Examinations and tests (2)	Requirement paragraph	Insp. Method paragraph	First Article Insp.	Quality Conformance Inspection
Materials inspection	3.2 thru 3.2.5	4.9.1		(6)
Visual and mechanical examinations:				
Design and construction	3.2 thru 3.12	4.9.2	(3)	
Packaging	5.0 thru 5.3	4.8	(4)	N/A
Operating environment tests:				
Temperature	3.4.2	4.9.11	(4)	N/A
Humidity	3.4.3	4.9.12	(4)	N/A
Electrical and mechanical tests:				
Resolution of fine detail	3.3.13	4.9.3	(5)	
Viewing surface brightness	3.3.14	4.9.4	(5)	
Viewing surface contrast	3.3.15	4.9.4	(5)	
Carrier Durability	3.3.16.5	4.9.5	(4)	N/A
Acoustic noise	3.3.18	4.9.6	(5)	N/A
Electromagnetic interference	3.3.19	4.9.7	(5)	N/A
Shock	3.3.20	4.9.8	(4)	N/A
Vibration	3.3.21	4.9.9	(4)	N/A
Collapsibility	3.3.22	4.9.10	(4)	
Film plane temperature	3.5	4.9.13	(5)	N/A
Viewer operating test	3.6	4.9.14	(5)	

NOTES:

(1) All inspections are applicable to the first article inspection and all inspections, except those marked N/A, are applicable to the quality conformance inspections.
(2) Unless otherwise authorized by the contracting officer, inspections are to be performed in the sequence listed.
(3) Two units shall be subjected to this test.
(4) One unit shall be subjected to this test.
(5) One unit to be subjected to these tests with 60 Hz electrical input power and one unit with 50 Hz electrical input power.
(6) Materials inspection should be conducted as an occasional quality conformance inspection on a random basis to verify that materials being used in production conform to the specified requirements.

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4.4.1 Inspection procedure. When listed on a Contract Data Requirements List (CDRL)(DD Form 1423), a first article inspection procedure shall be prepared and submitted to the Government for approval (see 6.3).

4.4.2 Inspection report. When listed on a CDRL, a first article inspection report shall be prepared and submitted for approval (see 6.3).

4.5 Quality conformance inspection. Inspection of product for delivery shall consist of the quality conformance inspections identified in Table II.

4.6 In-process inspections. First article and quality conformance inspection shall be performed at appropriate intermediate assembly steps to determine compliance with all requirements which cannot be determined on the fabricated end item. The procedures and results of in-process inspections shall be documented.

4.7 Maintenance during inspections. The only maintenance permitted during first article and quality conformance inspections is the replacement of lamps or fuses that have opened because of a part failure.

4.8 Inspection of packaging. The sampling and inspection of Level C or industrial/commercial packaging shall be as specified in the acquisition document (see 6.2). Otherwise, the sampling and inspection of the preservation, packing, and interior package marking shall be in accordance with the groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.9 Methods of inspection.

4.9.1 Materials inspection. Materials inspection shall consist of certifications supported by verifying data, including test results where applicable, showing that the materials used in fabricating viewers are in accordance with the applicable requirements.

4.9.2 Visual and mechanical examination. The viewer shall be examined to verify that materials, design, and construction are in accordance with all specified requirements.

4.9.3 Resolution of fine detail (see 3.3.13). Using the viewer's lower magnification lens (18X or 24X), focus the viewer on a microimage on a 24:1 resolution test microfiche (see 6.5.17). Using the pattern definition method (see 6.5.13), visually resolve pattern chart 3.6 at the center and the four corners of the screen without refocusing. Using the viewer's higher magnification lens (36X or 48X), focus the viewer on a microimage of a 48:1 resolution test microfiche. Using the pattern definition method, visually

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resolve pattern chart 3.2 at the center and the four corners of the screen without refocusing. Compromise focusing may be used but after initial focusing, the resolution requirements must be met for microimages in the approximate center and all four corners of the resolution test microfiche.

4.9.4 Viewing surface brightness (see 3.3.14) and screen image contrast (see 3.3.15). The viewing surface brightness and image contrast shall be tested in accordance with the methods prescribed in ANSI/HMA MS12-1977. Distances to the eyepoint shall be measured from the center of the apparent screen. All tests shall be performed with viewers set for full brightness, with lenses and platens (when used) in the optical path, with transparent shields (when used) in the visual path, and with the input voltage maintained between 103.5 and 103.6 v RMS. The tests will be conducted with the 36X or 48X lens in the optical path and repeated with the 18X or 24X lens in the optical path. During first article testing, these tests will be repeated with the meter position (eye point) at a distance of 30 in. from the apparent screen, and, in addition, one test will be conducted with each lens in the optical path and with the brightness set on reduced brightness to determine that the reduced level of brightness at the center of the screen is within the limits prescribed in 3.3.14.1.

4.9.5 Carrier durability (see 3.3.16.5). The carrier installed in one first article sample unit shall be fully cycled for 8,000 times from right to left to right, 8,000 times from front to rear to front, and 8,000 times from load to view to load positions. The front-to-rear-to-front and load-to-view-to-load test cycles may be combined into a single set of cycles. Any failure shall stop the test. Carrier durability test shall considered acceptable when the entire sequence of tests are completed without any failure and the viewer conforms to the requirements of 3.3.16 through 3.3.16.4 and 3.6.

4.9.6 Acoustic noise (see 3.3.18). Using a sound level meter, conforming to the requirements of ANSI S1.4-1971, set on the A scale, measure the acoustic noise at a distance of 3 feet from each of the four sides and the top of the viewer. The measured noise level shall not exceed the specified values.

4.9.7 Electromagnetic interference (EMI) (see 3.3.19). The viewer shall be tested to comply with the requirements for Visual Displays installed in Surface Ships (Class A4) per MIL-STD-461 and MIL-STD-462.

4.9.8 Shock (see 3.3.20). The viewer shall be tested as an item of ground equipment in accordance with MIL-STD-810, Method 516, Procedure I. After testing the viewer shall conform to the requirements of 3.6.

4.9.9 Vibration (see 3.3.21). The viewer shall be tested in accordance with MIL-STD-167-1 (Ships), type I, and shall conform to the requirements of 3.6 both during and after testing.

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4.9.10 Collapsibility (see 3.3.22). The viewer shall be erected. The viewer shall then be completely collapsed and re-erected 4000 times. Any failure shall stop the test. Upon completion of testing, the viewer shall conform to the requirements of 3.6.

4.9.11 Temperature test (see 3.4.2). Test viewers in accordance with MIL-STD-810, method 501.2, procedure II, with the following changes:

Step 2: Set temperature at 105 degrees F. Proceed to steps 7 through 9.
Step 8: Open temperature chamber, as necessary, and rapidly conduct an operational test in accordance with MIL-V-80240, 4.9.14.
Step 10: Set temperature at 35 degrees F. Repeat steps 7 through 9 then proceed to steps 15 through 17.
Step 15: Remove the test unit from the test chamber and allow it to stabilize at ambient temperature.
Step 17: Review the test data to determine whether or not the test unit conformed to the requirements of MIL-V-80240.

4.9.12 Humidity (see 3.4.3). The viewers shall be tested in accordance with MIL-STD-810, method 507.2, as a non-hazardous item of equipment using procedure I and cycle 3 (natural cyclic high humidity environment) or procedure III and cycle 5 (aggravated induced variable high humidity environment) and using the test duration specified in MIL-STD-810, Table 507.2-II. When visual examination is specified in the test procedure, the test unit shall also be given an operational test in accordance with MIL-V-80240, 4.9.14, to determine conformance to 3.6.

4.9.13 Film plane temperature (see 3.5). The viewers shall be tested in accordance with ANSI PH3.46-1971, except ambient temperature shall be 105 degrees F and film density shall be 1.3 or greater.

4.9.14 Viewer operating test (see 3.6). The viewers shall be connected to a power source and turned on. Each control shall be checked for correct operation. With a 24:1 print test microfiche in the carrier and the viewer's lower magnification lens in the optical path, the projection image on the viewing surface shall be fully enlarged to the limits required for the viewers. With a 48:1 print test microfiche in the carrier and the viewer's higher magnification lens in the optical path, the projection image on the viewing surface shall be fully enlarged to the limits required for the viewers. The viewing screen shall be examined for compliance with requirements herein. Any non-conformance shall be cause for rejection of the viewers (see 6.5.14).

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2 and 6.6).

5.1.1. Level A.

5.1.1.1 Cleaning. Viewers shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Viewers shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packs. Viewers shall be individually packed in accordance with MIL-P-116, submethod IA-14, insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. Hardware and accessories packages, such as maintenance manuals and parts accompanying each viewer, shall be conspicuously marked and placed inside the unit pack in a manner that will not damage the viewers or the accessories but will be obvious when the viewer is being unpacked to prevent their being discarded as packing material when reasonable care is exercised. The unit container shall conform to PPP-B-636.

5.1.1.5 Intermediate packing. Not required.

5.1.2 Level C. Preservation of viewers shall be in accordance with requirements specified in acquisition documents (see 6.2 and 6.6).

5.2 Packing. Packing shall be level A or C as specified (see 6.2 and 6.6). Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.2.1 Level A. The unit packed viewers shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with 50 mm (2 in.) minimum width tape, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only. Unit packs with outer containers conforming to PPP-B-636, class weather resistant, special requirements, require no further packing provided the requirements for closure, waterproofing and banding specified herein are met.

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5.2.2 Level C. Packing of viewers shall be in accordance with requirements specified in acquisition documents (see 6.2 and 6.6).

5.2.3 Unitized loads. Unitized loads shall be used whenever total quantities for shipment to one destination equal 1.13 cubic meters (40 cubic feet) or more. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.3.1 Level A. Viewers, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.3.2 Level C. Unless otherwise specified (see 6.2), viewers, packed as specified in 5.2.2, shall be unitized with pallets and caps of the type, size and kind commonly used for this purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to the marking required by the contract or purchase order, each unit package, exterior container and unitized load shall be marked with fragile labels, arrows and the word "UP" as specified in MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The viewers are used to read microimages on 24X and 48X microfiche and aperture cards. The viewers are intended for use aboard ships and at military field activities world-side where severe environmental conditions require a more rugged design than is available in commercial viewers and/or where other military peculiar requirements are essential.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Number, title, and date of this specification.
b. Identify item required by part identification number (see 1.3) or by size, direction of projection, lenses required (see 1.2.3), and/or other requirement.
c. Issue of DODISS and supplements thereto to be used to identify correct issue of referenced documents (see 2.1 and 2.2).
d. Issue of government documents that apply, if different from those listed in DoDISS.
e. Issue of non-government documents that apply and that are not listed in DoDISS.
f. Contract Data Requirements List (DD Form 1423) (see 6.3).
g. Method of classifying items, if different (see 1.2).

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h. Size classification of items, if different (see 1.2.1).
i. First article inspection information, when required (see 3.1, 4.4, 6.3, and 6.4).
j. Lens requirement, when other than standard lenses are required (see 3.3.11 through 3.3.11.2).
k. If partial disassembly is NOT allowed for removal of lenses (see 3.3.11.4).
l. Voltage and frequency for which delivered units are wired, if other than 115 volts and 60 Hertz (see 3.3.17).
m. When a plug conforming to MIL-C-28777, figure 2 (230 volt), is required on the power cord (see 3.3.17.4).
n. Technical manual requirements, when technical manuals are required (see 3.3.5.1, 3.3.6.2, 3.3.8, 3,3,10.2, 3.3.17, 3.8 and 6.3).
o. Activity responsible for inspections, if other than supplier (see 4.1).
p. Requirements for sampling and inspection of packaging, when Level C or industrial or Level C preservation and packing are specified (see 4.8 and 6.6).
q. Levels of preservation and packing required (see 5.1 and 5.2).
r. Requirements for preservation, packing, and marking, when Level C or industrial/commercial packaging is specified (see 5.1.2, 5.2.2, and 6.6).
s. Requirements for unitizing loads for Level C or industrial/commercial packing, if different (see 5.2.3.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are required. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4.1	DI-T-4901	First Article Inspection Procedure	---
4.4.2	DI-T-4902	First Article Inspection Report	---
3.8	DI-TMSS-80527	Manual, Technical, Operating and Maintenance Instructions	---

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited in the DOD Form 1423.

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6.4 First article inspection (see 3.1 and 4.4). When a first article inspection is required, the first article sample should consist of two preproduction or first production units. The contracting officer should include specific instructions in acquisition documents regarding the arrangements for inspections, the approval of first article test procedures and results, and the disposition of the first article sample. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending acquisition.

6.5 Key words and definitions.

6.5.1 Ambient light. The light surrounding the viewer; the general room illumination brightness level.

6.5.2 Apparent screen. The actual screen on a rear projection viewer; and, on a front projection viewer, the opening at the front of the viewer that is in the same location as the actual screen would be if the viewer were a rear projection viewer.

6.5.3 Frame, film. The area of a photographic film exposed to light in a camera during one exposure, regardless of whether or not this area is completely filled by the document image.

6.5.4 Frame, microfiche. One microimage and margin contained within a film frame.

6.5.5 Grid, microfiche. A defined array of horizontal and vertical lines which divide a microfiche into uniform spaces called frames. The grid defines the arrangement of the rows and columns of microimages. The grid lines are not normally printed on the microfiche.

6.5.6 Image. Reproduction of printed matter created by an optical system (i.e.: the enlargement of a microimage which can be projected on a screen for viewing).

6.5.7 Magnification. The linear ratio of the size of the image to that of the object when viewed through, or projected by, an optical instrument. The terms 18X, 24X, 36X, and 48X are magnifications which indicate that a linear dimension on the enlarged image is larger than the same dimension on the microimage by 18, 24, 36, and 48 times respectively.

6.5.9 Microfiche. A sheet of film containing multiple microimages in a grid pattern of rows and columns.

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6.5.10 Microform. A generic term for any form, either film or paper, which contains microimages.

6.5.11 Microimage. A unit of information, such as a page of text or a drawing, too small to be read without magnification.

6.5.12 Optical system. The combination of a light source, lenses, mirrors, and a screen used in illuminating, enlarging and displaying a microimage for viewing.

6.5.13 Pattern definition method. A method of determining resolution whereby resolution test chart patterns are visually examined. A pattern is resolved when the two groups of lines constituting the pattern can be distinguished in both directions. For more specific information see ANSI/NMA MS23-1983, paragraph 6.3.7.

6.5.14 Print test microfiche. Microfiche containing one or more microimages of full pages of text that have been reduced from an 8.5 by 11 in. original (for 24X magnification) and an 11 by 14 in. original (for 48X magnification) at reduction ratios of 24:1 + 0.4% and 48:1 + 0.2%, respectively. The microimages on the print test microfiche shall have a line density of not more than 0.2 and a background density of not less than 1.7.

6.5.15 Reduction ratio. The linear ratio of the size of an original document to that of a copy (or microimage) of the object when reduced through an optical instrument. The terms 24:1 and 48:1 are microform reduction ratios by which the linear dimensions on the original documents have been reduced on the microimage. For example, a linear dimension on a 24:1 microimage is 1/24 of the same dimension on the original document. Reduction ratio is comparable to magnification whereby a 24X magnification will restore a 24:1 microimage to the original document size.

6.5.16 Resolution. The ability of optical systems and photographic materials to make the fine detail of an object visible, expressed as number of discernible lines per millimeter.

6.5.17 Resolution test microfiche. Microfiche containing microimages that have been reduced from an 8.5 by 11 in. original (for 24X magnification) and an 11 by 14 in. original (for 48X magnification) at reduction ratios of 24:1 + 0.4% and 48:1 + 0.2% respectively. Each of the microfiche shall have a microimage in its center and each of its four corners. As a minimum each of the originals from which these microimages are made must have resolution targets in the center and the four corners and they must have a straight line of known length. The resolution targets must be comparable to the Microcopy Resolution Chart Number 1010a of the National Bureau of Standards, excluding the resolution patterns 1.0 through 1.8 and 10 through 18. One acceptable set of resolution test microfiche are the PSC Microfiche Test Charts MPT248.5NF and MPT4811NF, for 24X and 48X magnifications respectively, which are

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available from Photographic Sciences Corporation, 770 Basket Road, Webster, NY 14580-0338, at a cost of approximately $40.00 each (see 6.5.13).

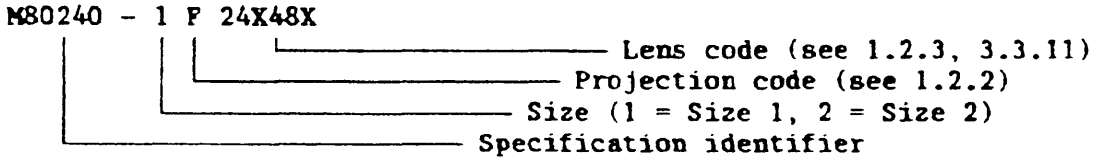
6.5.18 Screen. A surface onto which an image is projected.

6.5.19 Viewer. A projection device used for viewing a microimage with the unaided eye.

6.5.20 Viewing zone. All points that are within a 60 degree wide (apex angle) right circular cone whose apex is centered on the apparent screen, whose axis is perpendicular to the apparent screen, and whose length, along the axis, extends 36 in. in front of the apparent screen. At 26 in. away from the apparent screen, this conic zone will describe a 30 in. diameter circular disk that is parallel to and concentric with the apparent screen.

6.6 Army Level C preservation and packing. When Level C preservation and packing is required on an acquisitions by army activities, the contracting officers should specify that preservation and packing shall be in accordance with the requirements of MIL-STD-1190(SM).

6.7 Part or identifying number (PIN). The microfiche viewers covered by this specification are identified by a the following specification-based part or identifying number system. The initial part number characters "M80240" identify this specification as the applicable document for these items.



6.8 Metric conversion. This specification is prepared using inch-pound units based upon the yard and the pound commonly used in the United States of America and defined by the National Institute of Standards and Technology (NIST), formerly known as the National Bureau of Standards. Metric equivalents of inch-pound values may be derived utilizing FED-STD-376.

6.9 Changes from previous issue. Due to the extensiveness of the changes in this revision, no attempt has been made to identify changes from the previous issue.

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Custodians:

Army - CR
Navy - SA
Air Force - 07

Preparing activity:

DLA-GS

(Project 6730-0124)

Review activities:

Army - AL, AR, AT, AV, EA, MI, ME, SC, SM, TM
Navy - AS, CG, MC, NV, OS, SH, YD
Air Force - 70, 99
DOD - DO

User activities:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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3. DOCUMENT TITLE
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5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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8. PREPARING ACTIVITY

a. NAME Defense General Supply Center (GS)	b. TELEPHONE (Include Area Code) (1) Commercial (804)275-3900	(2) AUTOVON 695-3900
c. ADDRESS (Include Zip Code) DGSC-SSC Richmond, VA 23297-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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